Deakin University
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Sydney Housing Price Prediction and Decision Support System

SIT307 – Machine Learning

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Abstract

This project establishes a full pipeline, including data scraping, wrangling, modelling, and insights delivery. Predicts residential sale price from observable listing characteristics for three contrasting Sydney suburbs (Blacktown, Chatswood, Mosman), and serves the trained model through a small web–based application using `streamlit`. The implementation of this project can be found in this GitHub repo Sydney_housing.

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1 Problem Definition and Data Collection

1.1 Problem definition and suburb selection

An agency needs a tool estimating sale price from observable characteristics: supervised regression, predicting $y \in \mathbb{R}^+$ from features $\mathbf{x}$. The framing is hedonic [1, 2]. Accuracy alone is insufficient, because the agency also needs to know when to distrust a prediction, which motivates Section 4.

Three suburbs were chosen to span deliberately different markets, so location would be a first-order predictor. They differ in level, spread and stock (Table 1), i.e. Blacktown house-dominant and affordable, Chatswood unit-dominant, Mosman mixed and premium. Suburb and type are entangled by construction, so both are retained as inputs.

	Blacktown	**Chatswood**	**Mosman**
Retained sales n	79	62	47
Median price	\$950,000	\$1,417,500	\$1,760,000
Mean price	\$901,570	\$1,716,668	\$2,737,883
Std. deviation	\$334,345	\$970,869	\$2,116,702
Range	\$0.41–1.88M	\$0.70–4.80M	\$0.70–8.70M
Houses / Apartments	47 / 25	13 / 48	11 / 28
Other stock	6 Twn, 1 Vil	1 Twn	4 Twn, 2 Ret, 2 Semi

Table 1: Suburb contrast, computed in the notebook from the retained sample ($n = 188$). Twn = townhouse, Ret = retirement living, Semi = semi-detached.

1.2 Collection, quality and limitations

Records came from public `domain.com.au` sold listings [3], with context from Domain profiles [4] and ABS SEIFA [5]. realestate.com.au was excluded, as it denied programmatic access and REA Group has an enforcement record [6]. Every row carries `source_url` and `retrieved_at` for verification purpose.

The original collection comprises of 239 properties, with missing values present in different columns. Therefore, we aggressively discard incomplete rows, such that, there are more than 100 properties left with at least 30 properties in each suburb left. Having removed the incomplete rows, `area_sqm` is the only single source of missingness, so it alone was exempted. Its remaining 64 gaps (34%) are imputed inside the modelling pipeline. Admittedly, three of those six columns are not model inputs, and it cost 27 Mosman sales, contributing to the sparse Mosman tail in the price distribution. Cross-validating all three models on both the full and filtered sets confirms it is nonetheless the better choice, rather than keeping the missing values and imputing them in the modelling pipeline (notebook, *Keeping versus dropping incomplete rows*).

Noticeably, there are three apparent limitations. First, sales span only March–August 2026 with uneven coverage, so the temporal term cannot be extrapolated. Second, `area_sqm` remains 34% missing, so that column depends on imputation. Third, the SEIFA and `suburb_*` columns resolve to three distinct rows, restating the suburb label rather than adding property-level information.

We also identify a potential source of bias is selection on price disclosure. Collection filtered to listings showing a price, so withheld-price sales are absent by construction, and withholding is not random. The sample therefore under-represents the top of each distribution, most acutely in Mosman. Recovering those sales requires the NSW Valuer General's statutory register [7], which records every transfer regardless of advertised disclosure.

2 Data Understanding and Feature Engineering

2.1 Distribution, suburbs and time

Prices are heavily right-skewed, yet near-symmetric on a log axis (Figure 1). This implies the price effects are multiplicative, and thus, the price target is logarithmically scaled in the modelling pipeline. Dispersion grows with level, Mosman's standard deviation ($2.12M) is over six times Blacktown's ($334k); however, extreme values are a property of the market, not outliers.

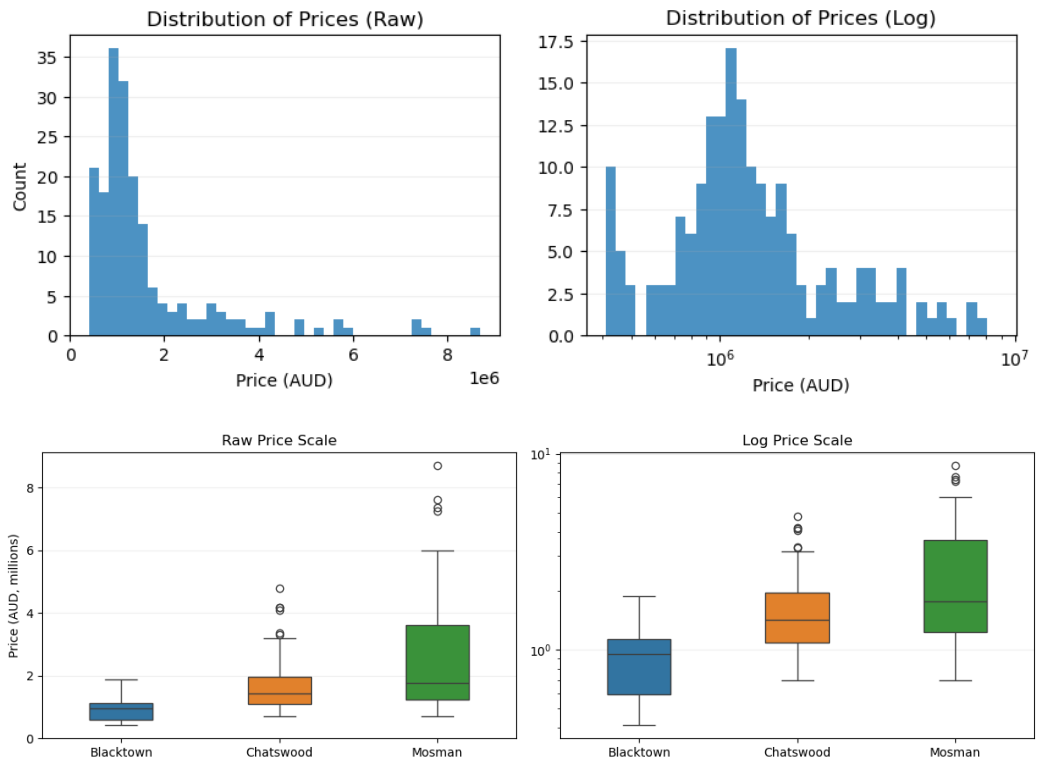


Figure 1: Price distribution in raw and log scale (top); by suburb, raw and log scale (bottom).

The temporal view is least trustworthy, across six months, badly unbalanced monthly coverage, and no repeated seasonal cycle (Figure 2). The correlation heatmap (Figure 3) shows the SEIFA block is redundant with suburb.

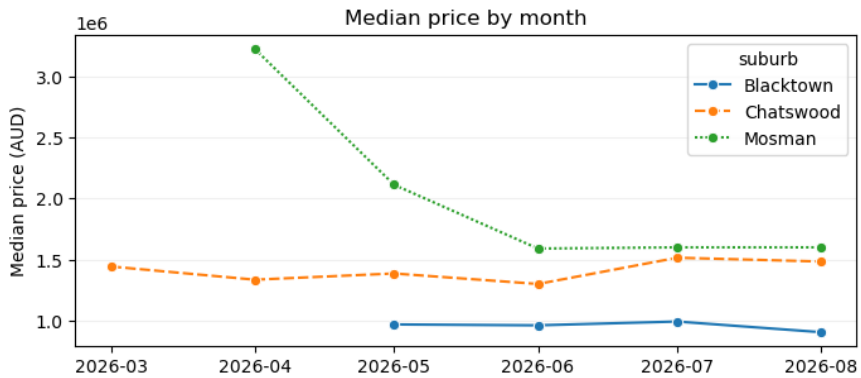


Figure 2: Median price by month. Coverage is uneven: Blacktown contributes no sales before May, Chatswood only two in August.

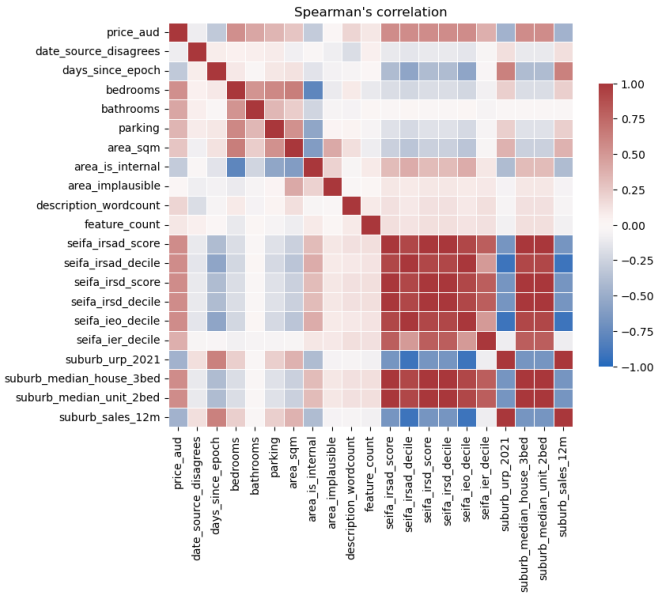


Figure 3: Spearman ρ correlation, target included. The ten SEIFA and `suburb_*` columns take only three distinct rows, one per suburb, and are mutually correlated at ≈ 1.

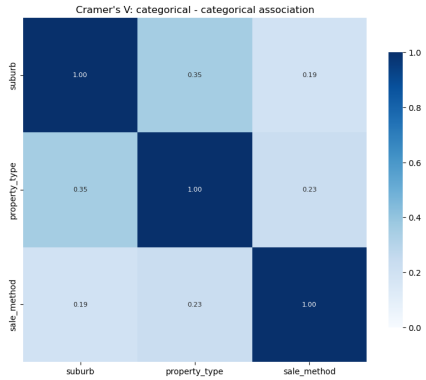


Figure 4: Cramer's V correlation, target excluded

2.2 Expected drivers and engineered features

Three variables were nominated beforehand as most influential, i.e. **suburb** (location dominated every comparison above), **property_type** (houses and units priced by different logic) and **price tercile** (to expose whether accuracy depends on position in the price range). From these, `is_unit` collapses six types into a land-versus-strata binary, avoiding near-empty categories such as Villa ($n = 1$); `bed_bath` and `beds_x_unit` summarise size and let a bedroom's value differ by type, `t` and `month` carry time, six `kw_*` flags recover condition signals from agent text.

Remarkably, only two of three survive as inputs, since the price tercile is derived from the target, which leaks a lot of information to the model. Thus, it serves diagnosis but is never available at prediction time, a limitation of the guess rather than the engineering. The third nominated driver, size, is measured two ways, i.e. directly by `area_sqm`, and indirectly by the room-count composites. The feature set therefore encodes the intended understanding (location first, type second, size third), though the direct size (`area_sqm`) measurement rests on imputation for a third of the sample.

3 Model Development and Evaluation

3.1 Models, pipeline and prior expectation

Three models represent different inductive biases. **Ridge**, regularised linear: with $n = 188$ and one-hot categoricals the design matrix is wide, and the L_2 penalty stabilises correlated inputs such as `bedrooms` and `bed_bath`. **Random Forest** bags deep trees, capturing interactions unprompted. **AdaBoost** fits residuals sequentially, chasing structure the others miss but most sensitive to atypical records.

All preprocessing sits inside the pipeline (median imputation, standardisation, and one-hot encoding with `min_frequency=3`), so it refits within every fold and cannot leak, which is important for `area_sqm`, whose 64 missing values are filled from each training fold's median. The target is log(price). Data is split 131/57 and tuned by grid search under 5-fold cross-validation on RMSE.

Prediction. The ensembles were expected to win, since a small dataset mixing categorical and numeric inputs across three different suburbs favours trees, more expressive power than Ridge.

3.2 Results and recommendation

Model	Best parameters	RMSE	MAPE%	MAE	R^2_{cv}	R^2_{tr}	gap
Ridge	`alpha`=0.231	**0.2055**	**16.78**	**0.1652**	0.9131	0.9358	**0.0227**
Random Forest	`depth`=4, `leaf`=3, `n`=100	0.2170	16.97	0.1705	0.9030	0.9411	0.0380
AdaBoost	`lr`=0.05, `n`=300	0.2257	18.15	0.1814	0.8951	0.9321	0.0370

Table 2: Out-of-fold results on the training split (RMSE, MAE and R^2 in log units; MAPE back-transformed to AUD). All three share identical folds.

From the table, no model underfits (all achieve $R^2_{cv} \geq 0.89$). Indeed, they do not overfit badly, since all gap are insignificant, at most 0.038 in `Random Forest`. Looking at the learning curves, it progressively falls when the fit size increases, which indicates that more training samples may improve performance.

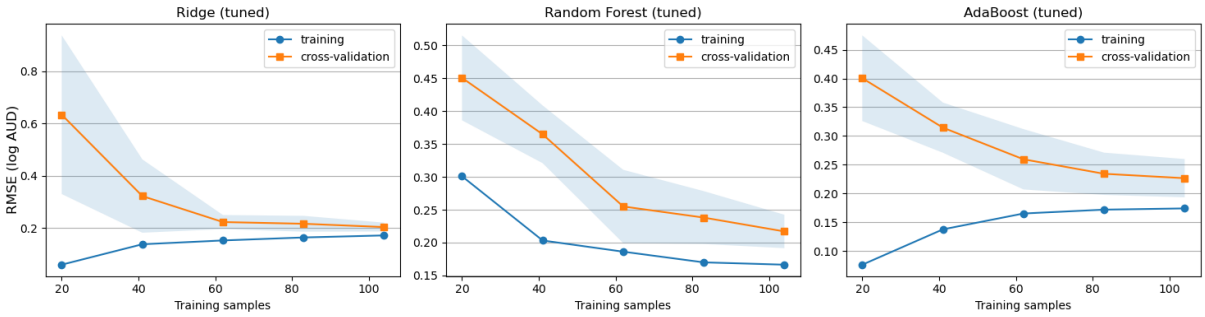


Figure 5: Learning curves. The cross-validation curve is still falling at the largest fit size (roughly 105 rows: four fifths of the 131 training rows) for all three models.

Model	Corr
`Ridge`	0.095
`Random Forest`	0.416
`AdaBoost`	0.330

Table 3: Pearson r correlation between each model's absolute log residual and the actual price.

Considering the table above, evidently, `Ridge` has errors that do not scale with price (correlation 0.095, against 0.416 and 0.330). This is the most decisive quantitative result driving our choice of best model. Let us present how each model produce residual errors.

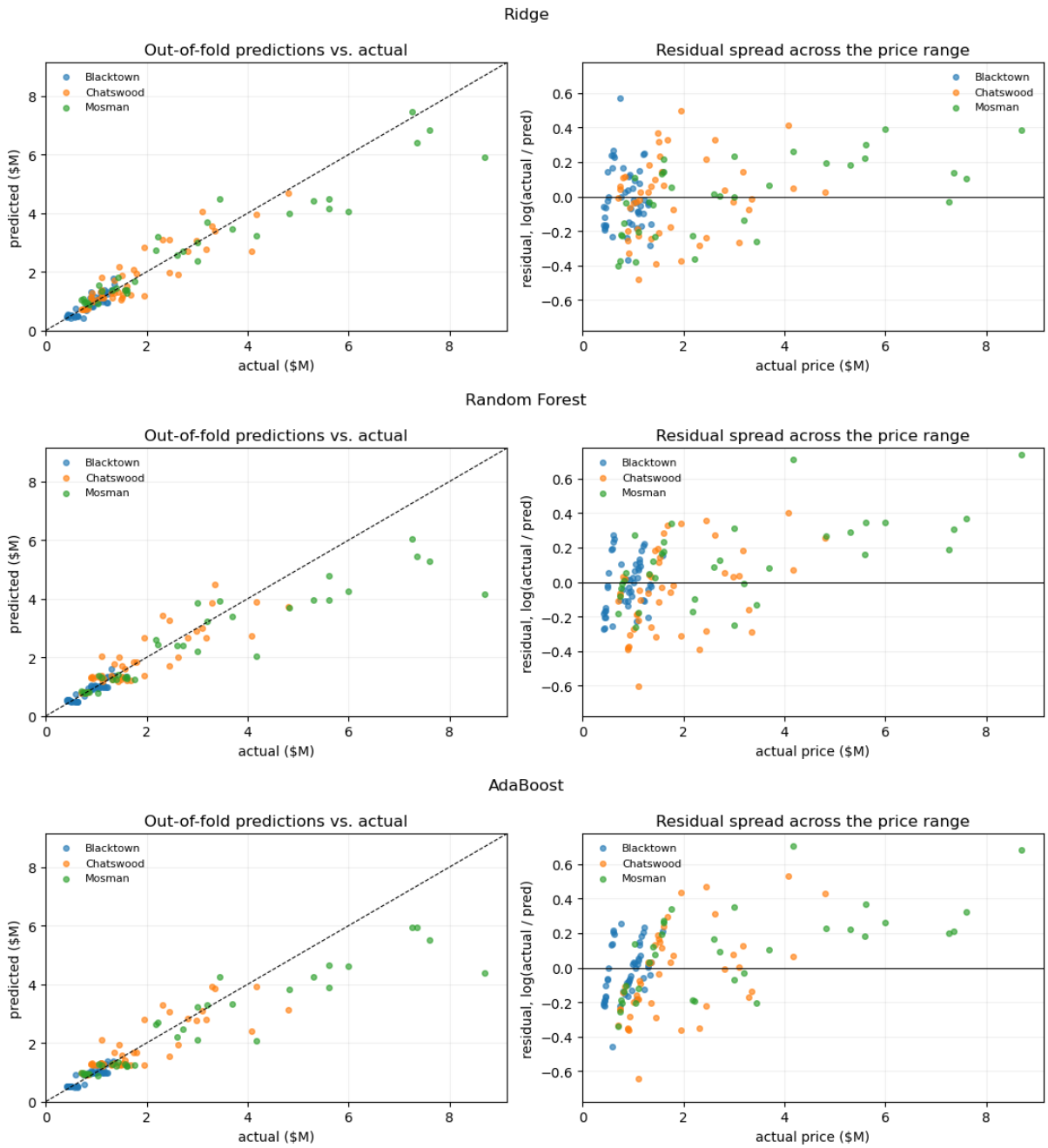


Figure 6: Out-of-fold predictions and residual spread, coloured by suburb: Ridge (top), Random Forest (middle) and AdaBoost (bottom).

The prior expectation was not supported. `Ridge` leads every cross-validated metric, is least over-fitted, and is the only model whose accuracy holds across the price range, which is more useful than one accurate only where data is dense. With `area_sqm` imputed for a third of rows, the usable signal is largely additive in log-price, so the ensembles' flexibility bought variance at $n = 188$. On the untouched test set, `Ridge` scores RMSE 0.2199, MAPE 17.51%, R^2 0.8096, or roughly ±25%. Being slightly worse than the cross-validated 0.2055 is the expected direction, and small enough to indicate tuning did not badly overfit the folds.

4 Investigating Prediction Failures

4.1 The five largest errors

Failures are ranked on log(actual/predicted), not dollars: |dollar error| largely restates price, whereas the log residual is scale-free and finds where the model is genuinely confused.

Suburb	Type	Bed	Bath	Actual	Predicted	Err (%)
Mosman	Apartment	1	1	$1,570,000	$845,911	+46.1
Chatswood	Apartment	3	3	$1,385,888	$2,365,823	−70.7
Chatswood	Apartment	2	2	$2,150,000	$1,300,159	+39.5
Blacktown	House	3	2	$1,000,000	$1,516,701	−51.7
Chatswood	Apartment	1	1	$1,030,000	$703,932	+31.7

Table 4: The five largest prediction errors on the held-out test set (Ridge).

Four of five are apartments, three sharing room counts with far cheaper units elsewhere. The worst case, a one-bedroom Mosman apartment at $1.57M against $846k predicted, is the clearest. In particular, room counts barely separate a 1-bed/1-bath unit from any other, and where `area_sqm` is missing it is replaced by the sample median, so the outlook justifying its price is invisible to the model. Five isolated cases could be outliers; whether they are is settled by asking if the same errors recur across a group.

Cut	Segment	n	Mean |err| (%)	Signed bias (%)
Suburb	Blacktown	29	11.7	−4.9
	Chatswood	16	25.6	−6.2
	Mosman	12	20.9	−0.5
Property type	Apartment	30	21.9	−4.3
	House	21	12.9	−6.0
Price tercile	Low	19	13.7	−10.8
	Mid	19	17.2	−4.1
	High	19	21.7	+1.9

Table 5: Test-set error by segment. Negative bias indicates over-prediction. Minor property types (townhouse, villa and retirement living; 6 sales combined) are omitted.

Chatswood averages 25.6% against Blacktown's 11.7% despite mid-range prices, so this is not merely a tail problem. Apartments average 21.9% against 12.9% for houses. Signed bias flips from −10.8% in the low tercile to +1.9% in the high. In conclusion, the model over-prices cheap homes and under-prices dear ones, regressing toward the mean and disadvantaging owners of unusual properties.

4.2 Limitations and when to distrust an estimate

Units are inherently harder to model *with these features*. Houses separate tolerably by room counts, whereas two apartments identical in bedrooms, bathrooms and parking can differ several-fold on floor level, aspect and outlook, none of which is recorded here. What would most improve performance is missing measurement, not a better algorithm.

Remarkably, estimates for one-bedroom units are least reliable, estimates at either price extreme are biased toward the middle, and any estimate outside March–August 2026 extrapolates the temporal term with no evidence.

5 Final Deployment and Reflection

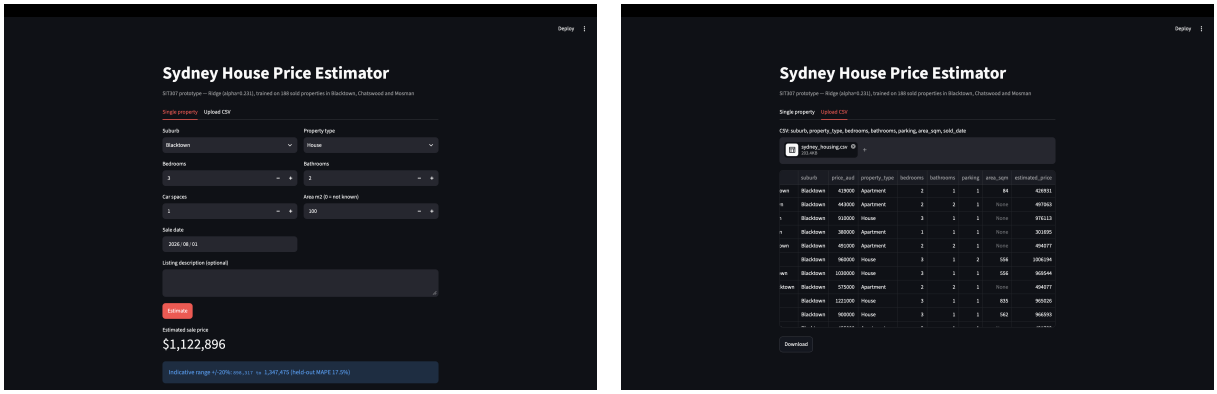
5.1 The application

The deployed model is the Ridge pipeline refit on all 188 sales. The split existed to obtain an unbiased error estimate, and once that is recorded there is no reason to discard 30% of the evidence. It is serialised to `src/model.joblib` with its column lists, so the app builds feature rows from the definitions used in training.

`src/app.py` is a Streamlit prototype loading that bundle, offering a single-property form and a CSV upload. Inputs are only fields readable off a listing, including floor or land area, which can be left blank. Derived terms are computed internally. The app exponentiates before display and reports a ±20% band rather than a point estimate.

```
pip install -r requirements.txt
jupyter nbconvert --to notebook --execute --inplace notebooks/lab.ipynb
streamlit run src/app.py        # from the project root; serves localhost:8501
```

The notebook must run first, since it writes `src/model.joblib`. Full instructions, including rebuilding the dataset from the cached raw pages, are in `README.md`.



(a) Single-property estimate. (b) Batch estimates from an uploaded CSV.

Figure 7: The Streamlit application. Estimates are shown with a ±20% band, and warnings fire on the cases Section 4 identified as unreliable.

5.2 Reflection

Data collection was the binding constraint, not modelling. 188 of 239 listings survived the completeness rule, and every later limitation traces to that number. The learning curves never flattened, so more sales beat a better estimator. Which rows to drop was itself a modelling decision: exempting `area_sqm` from the completeness rule preserved the suburb balance, at the

cost of imputing it for a third of the sample.

Evaluation choices changed conclusions. Scoring in-sample rather than out-of-fold reversed the ranking; MAPE on the logged target read 1.15% instead of 16.4%. Neither raises an error, and both would have been reported as findings. The target transform also mattered more than the algorithm, since logging moved Ridge from last to first.

Accuracy against deployability. The most direct signals were hardest to serve. `area_sqm` is a third missing, so the application must accept a blank and fall back on the training median, and the `kw_*` flags proved near-null. The deployed model asks only for fields readable off a listing in seconds, accepting ±25% error for an interface needing no preparation.

Bias and fairness. Three suburbs are not Sydney, and coverage within them is uneven, at 79 Blacktown sales against 47 Mosman, with error following at 11.7% against 25.6% in Chatswood. Since suburb is an explicit input and SEIFA proved to be suburb restated, the model encodes existing spatial inequality and would reproduce it in any decision it informed.

Given more resources. A longer window would separate seasonality from trend and fill the sparse Mosman tail; capturing `area_sqm` at collection would remove the need to impute a third of it; nested cross-validation would remove the optimism left in the CV figures.

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